Generative Relational Coherence

Toward a Higher-Level Ontology of Physics

Abstract

The representational fragmentation exposed by the Yang–Mills mass-gap problem points beyond a philosophy of mathematical description toward a possible ontology of physics. The appropriate generalization is not that the universe is simply information. Information presupposes distinguishable alternatives, physical relations, constraints, probabilities, and rules governing how distinctions propagate. A more fundamental proposal is *generative relational coherence*: the universe is a dynamically self-consistent structure of relations and possible processes in which local constraints generate stable global invariants across scale, context, and representation. Fields, particles, regulators, renormalization, continuum limits, and mass gaps can then be understood as different kinds of presentation, transformation, coherence, and rigidity within this structure. This essay develops that proposal and offers a precise answer to the question of what a universe is from this point of view.

1 From the Yang–Mills problem to a general ontology

The conceptual statement arising from the Yang–Mills mass-gap problem is:

> The Yang–Mills mass-gap problem is fundamentally the problem of generating and transporting infrared spectral rigidity across a chain of representations whose natural variables separately expose locality, gauge invariance, positivity, scale dependence, and spectral structure.

This formulation can be generalized:

> Fundamental physical problems concern how local relational constraints generate global forms of rigidity, and how those rigidities remain identifiable across representations that separately expose different aspects of physical reality.

Schematically,

$$\boxed{\text{local relational rules} \longrightarrow \text{collective organization} \longrightarrow \text{global invariant rigidity.}} \tag{1}$$

Each arrow can involve the composition of many degrees of freedom, changes of scale, elimination of redundant variables, singular limits, emergence of new effective objects, and translation between mathematical representations.

For Yang–Mills theory the pattern becomes

$$\text{local non-Abelian interaction} \longrightarrow \text{organized quantum vacuum} \longrightarrow \text{infrared spectral rigidity.} \tag{2}$$

Related patterns appear when microscopic reversibility generates macroscopic irreversibility, local quantum correlations generate topological order, microscopic interactions generate thermodynamic phases, local causal processes generate effective spacetime geometry, or quantum many-body dynamics generates quasiparticles.

The higher proposition is therefore:

> Physics studies the generation, transformation, and preservation of relational rigidity.

2 What is more fundamental than information?

The statement that the universe is information is suggestive but incomplete. Information already presupposes a space of alternatives, criteria of distinguishability, physical observables, probability assignments, and rules relating one subsystem to another. A more fundamental hierarchy is

$$\boxed{\text{relations} \longrightarrow \text{constraints} \longrightarrow \text{distinctions} \longrightarrow \text{information} \longrightarrow \text{stable objects.}} \tag{3}$$

2.1 Relations

Relations express how physical possibilities depend on one another: causal influence, spatial adjacency, interaction, entanglement, correlation, composition, and exclusion.

2.2 Constraints

Constraints determine which joint relational configurations are physically possible. Dynamical equations, conservation laws, causal order, positivity, symmetry, gauge constraints, and composition rules belong to this level.

2.3 Distinctions

A distinction is a physically realizable contrast between possibilities. It does not require a conscious observer. A detector, field, environment, or second subsystem may distinguish two physical states through interaction.

2.4 Information

Information quantifies distinguishability within a physical context. If ω and ν are states available in a context c, one may write schematically

$$D_c(\omega, \nu) = \sup_{O \in \mathcal{A}(c)} |\omega(O) - \nu(O)| \,, \tag{4}$$

where $\mathcal{A}(c)$ is the class of observables accessible in that context. The available information depends on the physically realizable distinctions, not merely on an abstract list of hidden values.

2.5 Stable objects

An object is a pattern of relational structure that remains identifiable under an appropriate class of transformations. Information is therefore not an independent substance from which relations are built. It is a quantitative measure defined on a prior structure of constrained relational possibilities.

3 Objects as invariants of transformation

Within this ontology, an object is not fundamentally a self-contained bearer of intrinsic properties. It is a stable structure persisting across transformations of description, scale, localization, and interaction.

A particle, for example, can appear as

- a pole or characteristic singularity of a correlation function;

- an irreducible representation of spacetime symmetries;
- a spectral sector of a Hamiltonian;
- a localized detector response;
- an asymptotic scattering state;
- an excitation of an underlying field.

None of these manifestations alone is simply the particle. Where the relevant equivalence and reconstruction theorems apply, the particle is the coherent invariant relating them.

This suggests the definition

$$\boxed{\text{physical object} = \text{stable equivalence class of relational manifestations.}} \tag{5}$$

Different entities exhibit different forms of stability.

Entity	**Characteristic form of stability**
Particle	Spectral and symmetry stability
Field	Local response and compositional stability
Phase of matter	Stability under perturbation and coarse-graining
Topological excitation	Stability under local deformation
Spacetime event	Stability within causal and localization relations
Conservation law	Invariance under admissible dynamics
Mass gap	Stability of the vacuum against arbitrarily low-energy excitation

Table 1: Physical entities as distinct forms of relational stability.

Objects can consequently be real without possessing one privileged mathematical presentation.

4 Reinterpreting the Yang–Mills statements

The earlier propositions can now be given an ontological reading.

4.1 Gauge fields are coordinates on physical information

A gauge field A_μ is not directly physical because gauge-related configurations represent the same physical situation. Its role resembles a coordinate system on an underlying relational structure. The physical content is carried by such structures as curvature, holonomies, Wilson loops, gauge-invariant correlations, responses to admissible probes, and the physical state space.

Thus, schematically,

$$A_\mu \quad \text{is a presentation, while} \quad [A_\mu]_{\text{gauge}} \quad \text{carries physical identity.} \tag{6}$$

Even the gauge-equivalence class need not be the final ontology. It may itself be a representation of a more directly observable-relational structure.

4.2 Regulators are temporary realizations

A lattice is more than a computational approximation. It is a concrete realization of a family of relational constraints. The continuum theory is physical only if a coherent structure survives as

$$a \longrightarrow 0, \qquad L \longrightarrow \infty. \tag{7}$$

The lattice variables may disappear while their invariant observable relations converge. The regulator is therefore a temporary carrier of physical structure.

4.3 Particles are spectral interpretations

A particle need not be a microscopic primitive. It can be a stable spectral organization of the theory:

$$\mathrm{Spec}(H) = \{0\} \cup \{\text{isolated massive sectors}\} \cup \{\text{spectral continua}\}. \tag{8}$$

In a strongly interacting theory the natural physical excitations may be very different from the elementary variables in the classical action. In pure Yang–Mills theory, gauge potentials appear in the local equations, whereas gauge-invariant composite excitations are expected in the physical spectrum.

4.4 Renormalization transforms descriptive scale

Renormalization does not ordinarily represent physical time evolution. It changes which distinctions are retained at a selected resolution. A coarse-graining transformation

$$\mathcal{R}_{\ell \to \ell'} : \mathfrak{R}_\ell \longrightarrow \mathfrak{R}_{\ell'} \tag{9}$$

forgets some microscopic distinctions while preserving selected long-distance predictions. Universal quantities are those that remain stable despite this loss of microscopic descriptive detail.

4.5 The continuum is a coherence condition

The continuum need not be understood merely as an infinitely fine lattice. It is the coherent limiting object represented by a compatible family of regulated descriptions. If ω_a is the state at cutoff a, then for appropriate observables one requires

$$\lim_{a \to 0} \omega_a(O_a) = \omega(O), \tag{10}$$

together with compatibility across observable products, symmetries, localization, positivity, and scale transformation. The continuum is thus a consistency structure across a family of approximations rather than another microscopic picture.

4.6 The mass gap is infrared rigidity

The mass gap states that the vacuum cannot be changed into a distinct physical excitation at arbitrarily small positive energetic cost:

$$H \geq \Delta(1 - P_0), \qquad \Delta > 0, \tag{11}$$

where P_0 projects onto the vacuum sector. Ontologically, this says that the vacuum is rigid against sufficiently long-wavelength physical deformation.

The corresponding decay

$$\langle O(x)O(0)\rangle_c \sim e^{-\Delta|x|} \tag{12}$$

means that local physical distinctions do not propagate as indefinitely persistent low-energy correlations. The vacuum has a finite relational memory length.

5 The universe as a coherent family of contexts

The ontology can be made more precise by introducing physical contexts. Let

$$c = (\Omega, \ell, B, P, R), \tag{13}$$

where Ω is a spacetime region, ℓ a scale or resolution, B a set of boundary conditions, P a class of physical probes, and R a mathematical presentation.

To every context c, assign

$$\mathfrak{U}(c) = \big(\mathcal{A}(c), \mathcal{S}(c), \mathcal{P}(c), \mathcal{C}(c)\big), \tag{14}$$

where

- $\mathcal{A}(c)$ is the observable algebra;
- $\mathcal{S}(c)$ is the state space;
- $\mathcal{P}(c)$ is the space of possible physical processes;
- $\mathcal{C}(c)$ records causal, compositional, and compatibility constraints.

Contexts are related by transformations

$$\Phi_{c\to d} : \mathfrak{U}(c) \longrightarrow \mathfrak{U}(d). \tag{15}$$

These transformations may express localization, inclusion of regions, change of resolution, coarse-graining, gauge quotient, change of coordinates, duality, continuum reconstruction, or physical time evolution.

A universe is not any single description $\mathfrak{U}(c)$. It is the structured totality

$$\mathbb{U} = \left(\{\mathfrak{U}(c)\}_{c\in\mathsf{C}}, \{\Phi_{c\to d}\}, \{\eta_{c,d,e}\}\right), \tag{16}$$

where $\eta_{c,d,e}$ expresses coherence between alternative transformation paths. For example, transforming from c to e directly should agree, exactly or within a controlled approximation, with transforming through d:

$$\Phi_{c\to e} \simeq \Phi_{d\to e} \circ \Phi_{c\to d}. \tag{17}$$

The universe is not one member of this family. It is the coherent organization of the entire family.

6 Global coherence without predetermined values

This ontology does not say that every observable has a predetermined value or that the universe is a hidden classical database. Quantum theory obstructs such a naive global value assignment. The compatible global object is instead a state or probability structure assigning expectations to physically admissible observables:

$$\omega_c : \mathcal{A}(c) \longrightarrow \mathbb{C}. \tag{18}$$

Compatibility requires, schematically,

$$\omega_d(O_d) = \omega_c(\Phi^*_{c\to d} O_d) \tag{19}$$

whenever c and d are exact descriptions of the same physical content. The universe may therefore be globally coherent without being a catalogue of simultaneously definite local facts.

7 Objectivity as certified coherence

The contextual character of representation does not entail relativism. Different representations are not automatically equally adequate. A representation can fail because it loses positivity, changes the physical observable algebra, fails to reconstruct causal locality, eliminates nontrivial interactions, has the wrong continuum limit, destroys spectral structure, or introduces an uncontrolled approximation.

Objectivity is therefore not independence from every representation. It is invariance under the admissible transformations connecting valid representations:

$$\boxed{\text{objective physical content} = \text{content stable under certified changes of representation.}} \tag{20}$$

This is a form of structural realism with a generative component. Reality contains both invariant structures and lawful processes through which new stable structures emerge.

8 Why transformations must remain typed

Not every transformation represents physical change. The ontology must keep distinct transformations distinct.

Transformation	**Physical or representational meaning**
Gauge transformation	Same physical state, different presentation
Coordinate transformation	Same structure, different coordinates
Exact duality	Same physical content, different ontology of variables
Coarse-graining	Reduced resolution and controlled information loss
Restriction to a region	Partial physical perspective
Continuum limit	Construction of a limiting theory
Time evolution	Actual physical change of state
Measurement interaction	Physical correlation and outcome formation

Table 2: Different transformation types must not be conflated.

Representational dynamics is a meta-level theory of how descriptions transform. Physical dynamics is one distinguished class of transformations within it. A precise formalization may

therefore require a double or higher categorical structure: vertical arrows for equivalent descriptions, horizontal arrows for restriction, coarse-graining, and physical processes, and higher cells certifying that different transformation paths are coherent.

9 Laws as coherence constraints

In an object-centered ontology, laws primarily describe how things move or change. In a relational-coherence ontology, laws have a broader function. They determine

- which local relational configurations are admissible;
- how systems and processes compose;
- which transformations count as physically equivalent;
- how descriptions at different scales correspond;
- which quantities are conserved;
- which emergent structures remain stable;
- how local information constrains global possibilities.

A physical law is therefore not merely a differential equation. It is a coherence condition across a family of possible physical contexts. The Yang–Mills action remains essential, but it is one compact local generator within a larger structure including gauge equivalence, observables, states, renormalization, positivity, limiting constructions, and spectral consequences.

10 What is a universe from this point of view?

A concise answer is:

> A universe is a dynamically self-contained and globally coherent system of relational possibilities, physical constraints, and composable processes, whose valid descriptions form a compatible family across localization, scale, and representation, and whose objects are the stable invariants generated within that family.

Schematically,

$$\boxed{\text{Universe} = \text{relations} + \text{constraints} + \text{processes} + \text{coherence} + \text{emergent rigidity}.} \tag{21}$$

From this perspective,

- spacetime is the organization of localization and causal composability;
- matter is a family of stable excitation structures;
- particles are persistent spectral patterns;
- fields encode possible local responses and correlations;
- symmetries classify equivalence and conservation;

- information measures physically available distinctions;
- laws regulate composition and compatibility;
- renormalization relates descriptions at different resolutions;
- objects are structures stable under relevant transformations;
- emergence is the generation of new invariants under composition and scale change;
- the universe is the maximal coherent physical process containing these relations.

11 The strongest generalized proposition

The original Yang–Mills statement can finally be generalized as follows:

> A physical universe is a generative relational structure in which local constraints produce global invariant rigidities, while lawful transformations connect the partial representations through which those rigidities become observable, calculable, and provable.

The Yang–Mills mass gap is a particularly pure test case. It asks whether a local, gauge-redundant, scale-free presentation generates a gauge-invariant infrared rigidity that survives every required transformation into the continuum physical theory.

This is why the problem is philosophically central. It tests whether physics can explain not merely the behavior of already constituted objects, but the emergence of stable physical objecthood itself from relational quantum structure.